Policy & Strategy Lab

Instructor Guide, Week 1: Define and structure

This guide covers Days 1 to 5. Each day follows the same pattern: an overview, the session plan, preparation, a slide-by-slide teaching narrative that matches the day's PowerPoint, likely misconceptions, discussion questions with what strong answers include, full facilitation notes for the activity, the workbook application, assessment, readings and the transition to the next day.

The running case in every demonstration is Madhyapur, a stylised city. Its figures are illustrative and must never be presented as real data. Rules, schemes and institutions named alongside it (the Solid Waste Management Rules, 2016, Swachh Bharat Mission (Urban) 2.0, the Seventh and Twelfth Schedules) are real; check current details before teaching.

Day 1: Problems, clients and decisions

Item	Detail
Week	1: Define and structure (Monday)
Central question	What does it mean to solve a public problem systematically, and who are we solving it for?
Session outcome	Each participant has sorted statements about your candidate cases into problems, symptoms, causes and solutions, and named a provisional decision-making role for each.
Competencies	A1, A3, A4
Anchor tool	Problem, symptom, cause and solution sort
Supporting tools (demonstrated only)	The seven steps and the eightfold path compared; Naming the decision-maker
Theory link	Rittel and Webber: Whose definition of the problem are you adopting, and what would count as solved?
Gate	No gate today.

Learning objectives

1. Explain how the Lab works: one case, one shared method, four gates.
2. Distinguish a problem from a symptom, a cause and a solution.
3. Name a decision-making role and the decision it faces.
4. Explain why public problems are harder to define than a firm's.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 25	How the Lab works	15 minutes
25 to 45	Problem, symptom, cause, solution	20 minutes
45 to 80	Activity: ten-minute challenge	35 minutes
80 to 105	Own case: candidate sort	25 minutes
105 to 120	Debrief and exit task	15 minutes

Preparation and materials

- The Day 1 slide deck (03 Slides)
- Worksheet 1.1 in Workbook 1
- Scenario card
- A4 sheets or one shared slide per pod
- Four colours of pen or highlight, one per category

Indian policy context

Welfare access in Indian districts: many schemes exist, yet the exclusion of eligible families is a recurring concern in audits and research. The problem is easy to misstate as a lack of awareness among citizens.

Teaching narrative, slide by slide

Slide 1. Title slide. Welcome everyone and introduce yourself in under a minute. Read the central question aloud. Tell participants that by the end of the session each of them will have sorted their own candidate cases and named who could act on them. Do not run long introductions: the opening problem starts the work immediately.

Slide 2. Today. Session outcome: by the end of the session each participant has sorted statements about your candidate cases into problems, symptoms, causes and solutions, and named a provisional decision-making role for each.

Slide 3. Opening problem. Give exactly two minutes of silent writing. Then ask three participants to read their sentence aloud and write them on the board without comment. Most will contain a cause (lack of awareness, corruption, poor internet) or a solution (a camp, an app). Do not correct them yet. You will return to these sentences after the next two slides.

Slide 4. How the Lab works. Keep this brief. The point to land is that the course is cumulative: nothing is done once and thrown away. Mention that all evidence will come from published and public sources; there are no interviews or surveys. Explain the pods and the four rotating roles (chair, sceptic, evidence checker, scribe) in one sentence; they start on Day 2.

Slide 5. The method: seven steps to a diagnosis, then a choice. This is the seven-step process associated with McKinsey and set out by Conn and McLean. Say plainly that it is strong on diagnosis and thin on choosing between options, which is why Week 4 borrows from policy analysis. Do not teach the steps in detail now; each will get its own day.

Slide 6. Four things that get confused with the problem. Introduce the spine case now. Madhyapur is a stylised municipal corporation of about a million people. The Solid Waste Management Rules, 2016 require households to segregate waste at source, yet in Madhyapur most hand over mixed waste. You will use this case for every demonstration in the course. Go back to the three opening sentences on the board and classify each phrase with the room.

Slide 7. Sort practice: six statements about Madhyapur. Reveal the categories one row at a time after the room has voted. Expect disagreement on the landfill statement (some will call it the problem). Use it to show that the choice of problem depends on the decision-maker: for a state environment department the landfill might be the problem; for the Commissioner deciding on collection, it is a consequence.

Slide 8. Every problem is defined for a decision-maker. The client in this course is a role, never a person you contact. Ask: who in the welfare scenario could act? Push past 'the government' to a role: the District Collector, the block development officer, the state department. Different roles need different analyses of the same issue. This is why the Case Approval Form on Day 2 asks for the role and the decision.

Slide 9. A thinker for your case. Rittel and Webber, "Dilemmas in a General Theory of Planning", Policy Sciences, 1973. They argued that most public problems are wicked: they have no definitive formulation and no stopping rule, and every attempt to solve one changes it. Put the question to the room: Whose definition of the problem are you adopting, and what would count as solved? Then ask each participant to answer it for their own case in one sentence.

Slide 10. Why public problems are harder to define than a firm's. This slide states the programme's central claim: the state is not a firm. Ask participants to add one difference from their own candidate case. Good additions: legal mandates, electoral cycles, frontline discretion, fixed scheme guidelines.

Slide 11. Activity: The ten-minute problem challenge. Do not intervene in the first ten minutes. Most pods will produce a cause or a solution presented as the problem, such as a lack of awareness among

beneficiaries. Read two or three first drafts aloud before the swap. In the debrief, show how a lack of awareness is a hypothesised cause, and ask what evidence would support or weaken it.

Slide 12. Your case: Worksheet 1.1. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Watch for participants who list only solutions; ask what they would observe if the problem were solved.

Slide 13. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Write your best candidate problem in one sentence, with no cause and no solution in it.

Slide 14. Before the next session. Finish Worksheet 1.1. For the candidate you are most likely to choose, open two public data sources and note what each covers, at what level (state, district or city) and for which year. Bring both to Day 2. About 45 minutes.

Likely misconceptions

Misconception	How to respond
A problem statement should explain why the problem happens.	Explanations come later, as hypotheses to test. A cause in the statement closes off the analysis before it starts.
Lack of awareness is the problem.	It is a candidate cause, and often a convenient one because it places the burden on citizens. Treat it as a hypothesis.
The decision-maker is the government.	Government is many offices with different powers. Name the role that can actually act.
Consulting methods do not apply to public problems.	They apply, with adjustments for contested goals, divided authority and legitimacy. Those adjustments are what the Lab teaches.

Discussion questions

Question	What strong answers include
What is the difference between a symptom and a problem?	A symptom is an observable sign; the problem is the gap itself, defined for a group. Strong answers note that symptoms are often felt by a different group from the one experiencing the problem.
Why name the decision-maker on Day 1?	The decision sets the scope, the criteria and the evidence that matters. Strong answers point out that two decision-makers would need two different analyses of the same issue.
Is wicked just another word for hard?	No. Rittel and Webber meant problems with no definitive formulation or stopping rule, where every intervention changes the problem. Strong answers give an Indian example and say what it implies for analysis.

Activity: The ten-minute problem challenge

Element	Detail
Purpose	Experience how quickly a group jumps to solutions, then rebuild the problem properly.
Scenario	A district runs more than twenty central and state welfare schemes. At a district grievance meeting, families from remote blocks say they receive nothing, although many appear eligible. The District Collector turns to your team: what is the problem, and what do we need to find out first? You have ten minutes.
Group size	Pods of four
Time	35 minutes
Materials	Scenario card; A4 sheets or one shared slide per pod; Four colours of pen or highlight, one per category

Element	Detail
Expected output	Before and after statements with a named decision-maker

Steps.

1. In ten minutes, write the problem in one sentence and list what you need to find out.
2. Swap with another pod. Mark every phrase as problem, symptom, cause or solution.
3. Rewrite your own statement so it contains no cause and no solution.
4. Name the role that would act on it and the decision that role faces.
5. Be ready to read your before and after statements aloud.

Facilitation. Do not intervene in the first ten minutes. Most pods will produce a cause or a solution presented as the problem, such as a lack of awareness among beneficiaries. Read two or three first drafts aloud before the swap. In the debrief, show how a lack of awareness is a hypothesised cause, and ask what evidence would support or weaken it.

Debrief questions.

- How many first drafts contained a cause or a solution?
- Which word in your first draft did the most damage?
- Did naming the decision-maker change what you thought you needed to find out?

Evaluation criteria	Common errors
The final statement contains no cause or solution	Treating a lack of awareness as the problem
The affected group and place are stated	Naming the government as the decision-maker
A role with authority is named	Listing information needs without priorities

Extension for strong groups. Ask the pod to write how a frontline worker, an excluded family and the District Collector would each state the problem.

Workbook application: Worksheet 1.1, Sorting your candidate cases

Separate what you believe about each candidate case into problems, symptoms, causes and solutions, and test whether anyone could act on it. Watch for participants who list only solutions; ask what they would observe if the problem were solved.

Hand in. Worksheet 1.1 with three sorted candidates and a provisional choice, by the start of Day 2.

Independent work

Finish Worksheet 1.1. For the candidate you are most likely to choose, open two public data sources and note what each covers, at what level (state, district or city) and for which year. Bring both to Day 2. About 45 minutes.

Assessment

Not graded. Checked at the start of Day 2 for completion and for causes or solutions hidden in problem statements (tracker code PF1).

Readings

- Essential: Rittel and Webber, "Dilemmas in a General Theory of Planning" (1973). Read the ten properties of wicked problems.

- Optional: Conn and McLean, Bulletproof Problem Solving (2018), chapter 1.
- Optional: Bardach and Patashnik, A Practical Guide for Policy Analysis, part I, step one: define the problem.

Close and transition

Exit task: Write your best candidate problem in one sentence, with no cause and no solution in it.
Transition: Tomorrow we turn your best candidate into a bounded, decision-focused problem statement and submit the Case Approval Form.

Day 2: Defining the problem

Item	Detail
Week	1: Define and structure (Tuesday)
Central question	How do we turn a broad concern into a problem statement precise enough to analyse and act on?
Session outcome	Each participant has completed the Problem Definition Canvas for your case and submitted the Case Approval Form.
Competencies	A2, A3, A4
Anchor tool	Problem Definition Canvas
Supporting tools (demonstrated only)	5W1H precision test; The case-narrowing ladder
Theory link	Amartya Sen: Are you measuring access to a resource, or what people are actually able to do with it?
Gate	No gate today.

Learning objectives

1. Write a problem statement that names population, place, institution, period and a measure of success.
2. Bound the scope: what is in, and what is deliberately left out.
3. Record competing framings and adopt one for a stated reason.
4. Check your case against the feasibility tests.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 30	Concept: precision and scope	20 minutes
30 to 50	Demonstration: canvas on Madhyapur	20 minutes
50 to 80	Activity: problem reframing lab	30 minutes
80 to 110	Own case: canvas and approval form	30 minutes
110 to 120	Debrief and exit task	10 minutes

Preparation and materials

- The Day 2 slide deck (03 Slides)
- Worksheet 1.2 in Workbook 1
- Four statement cards
- 5W1H prompt card
- Canvas template

Indian policy context

A household may hold a ration card and still be unable to draw its rations if the fair price shop is open only when the household cannot travel. A measure built only on card coverage would miss the problem.

Teaching narrative, slide by slide

Slide 1. Title slide. Return Day 1 exit tasks briefly: read two strong statements aloud. Today is about precision. By the end of the session every participant submits a Case Approval Form.

Slide 2. Today. Session outcome: by the end of the session each participant has completed the Problem Definition Canvas for your case and submitted the Case Approval Form.

Slide 3. Opening problem. Collect three rewrites. Point out which of the five questions each one answered. Most will narrow the place and the outcome but leave the decision-maker out. Say that the canvas you are about to see forces all of them at once.

Slide 4. The Problem Definition Canvas. Walk through the boxes in order. Emphasise two of them. Current condition needs a source and year, even if the source is imperfect; an unsourced number is worse than an honest description. Out of scope is a decision, not an omission: writing it down stops the case drifting in Week 3.

Slide 5. 5W1H turns a vague statement into a precise one. The Solid Waste Management Rules, 2016 place a duty on waste generators to segregate waste at source into separate streams. In the stylised Madhyapur case the evidence base is the corporation's own records; in a real case participants would cite the specific report and year. Ask which row was hardest to make precise in the opening exercise.

Slide 6. Madhyapur on the canvas. This is the spine case's canvas, and the standard for Gate 1. Note the illustrative label on the figure: participants must never invent numbers for their own cases. Point to the out-of-scope box: excluding landfill remediation is defensible because it has a different decision-maker and budget.

Slide 7. Narrowing a case: the ladder. Ask participants where their own candidate currently sits on the ladder. Anyone at rung one or two must climb today. The final rung is what makes the case analysable in four weeks.

Slide 8. A thinker for your case. Amartya Sen, Development as Freedom, 1999. Sen argued that development should be judged by people's capabilities, the real freedoms they have to do and be what they value, not only by the resources they hold. Put the question to the room: Are you measuring access to a resource, or what people are actually able to do with it? Then ask each participant to answer it for their own case in one sentence.

Slide 9. One problem, three framings. This is Rittel and Webber again: whose definition are you adopting? For the Commissioner's decision the compliance framing is natural, but a good analysis keeps the other framings in view, because they determine legitimacy and support later. Competency A3 asks for at least two framings to be recorded.

Slide 10. Is your case workable in four weeks? Five quick tests. These summarise the nine-criterion feasibility rubric on the Case Approval Form. A case needs at least 12 of 18, with no zero on evidence, institutional clarity or ethical safety. Tell participants that cases failing the evidence test are the most common problem, and the cheapest to fix today.

Slide 11. Activity: Problem reframing lab. Assign pod roles for the first time today: chair, sceptic, evidence checker and scribe. They rotate weekly. Challenge any statement that still contains poor, lack of or inadequate without saying compared with what. Finish with a five-minute gallery walk.

Slide 12. Your case: Worksheet 1.2. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Check that every participant has actually opened two data sources; if not, help them find one before they leave.

Slide 13. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Read your problem statement to your pod sceptic. Write down the one word they challenged.

Slide 14. Before the next session. Revise your canvas after the gallery walk. Find the published qualitative or field-based source the Case Approval Form asks for, such as an audit report, social audit, committee report or field study, and note its two most relevant findings. About 45 minutes.

Likely misconceptions

Misconception	How to respond
A precise statement must contain a statistic.	It needs a source for the current condition, but an unsourced number is worse than an honest description. Precision comes from naming the group, place, institution and gap.
A narrow scope means a trivial problem.	A narrow scope makes a serious problem analysable in four weeks. Breadth can return when discussing implications.
The desired condition is whatever I think is right.	Anchor it in a rule, a target, a norm, or an explicit value judgement stated openly.

Discussion questions

Question	What strong answers include
Why does the canvas ask for out-of-scope items?	Because scope is a choice with consequences; writing it down stops drift. Strong answers note that out-of-scope items can return as implications or limitations.
When does a measure of success miss what matters?	When it counts resources supplied rather than what people can do. Strong answers apply Sen's distinction to their own case.
What makes a decision-maker appropriate?	Authority to act on the problem within the timeframe, and a real choice to make. Strong answers check jurisdiction, not just seniority.

Activity: Problem reframing lab

Element	Detail
Purpose	Turn vague public concerns into precise, decision-focused statements, then test them against each other.
Scenario	Each pod receives four vague statements. Women do not work enough in India. Our cities have a waste crisis. Government schools are failing. Farmers do not get fair prices. Make each one usable by a real decision-maker.
Group size	Pods of four, one statement each
Time	30 minutes
Materials	Four statement cards; 5W1H prompt card; Canvas template
Expected output	Four rewritten statements; one posted for the gallery

Steps.

1. Take one statement each. Run it through 5W1H in five minutes.
2. Rewrite it with population, place, institution and period.
3. Name the role that could act and the decision it faces.
4. Pass it to the pod sceptic, who must find one remaining vagueness.
5. Agree the pod's best rewrite and post it for a gallery walk.

Facilitation. Assign pod roles for the first time today: chair, sceptic, evidence checker and scribe. They rotate weekly. Challenge any statement that still contains poor, lack of or inadequate without saying compared with what. Finish with a five-minute gallery walk.

Debrief questions.

- Which vague statement was hardest to make precise, and why?
- Did choosing a decision-maker force you to narrow the place?
- Where did the sceptic find vagueness you had not noticed?

Evaluation criteria	Common errors
Group, place, institution and period are named	Narrowing the place but leaving the group vague
No cause or solution in the statement	Replacing vagueness with an unsourced number
A role with authority is named	Choosing a decision-maker with no authority over the issue

Extension for strong groups. Write the same problem from the point of view of the most affected group and compare the two statements.

Workbook application: Worksheet 1.2, Problem Definition Canvas and Case Approval Form

Define your case precisely enough to analyse and act on, and confirm it is workable in four weeks. Check that every participant has actually opened two data sources; if not, help them find one before they leave.

Hand in. Canvas and Case Approval Form by the end of Day 2. Provisional approval is returned by Day 3.

Independent work

Revise your canvas after the gallery walk. Find the published qualitative or field-based source the Case Approval Form asks for, such as an audit report, social audit, committee report or field study, and note its two most relevant findings. About 45 minutes.

Assessment

Case Approval Form scored on the feasibility rubric: nine criteria from 0 to 2, at least 12 of 18, with no zero on evidence, institutional clarity or ethical safety. Provisional approval by Day 3.

Readings

- Essential: Bardach and Patashnik, A Practical Guide for Policy Analysis, part I, step one: define the problem.
- Optional: Sen, Development as Freedom (1999), introduction and chapter 1.
- Optional: PRS Legislative Research: browse the bill and budget summaries for your sector.

Close and transition

Exit task: Read your problem statement to your pod sceptic. Write down the one word they challenged. Transition: Tomorrow we break your problem into a structured set of questions with an issue tree, and you seal your first working answer.

Day 3: Structuring with issue trees

Item	Detail
Week	1: Define and structure (Wednesday)
Central question	How do we break a complex problem into questions we can actually answer?
Session outcome	Each participant has built a three-level MECE issue tree for your case, written and sealed a working answer, and started your Analyst's Ledger.
Competencies	B1, B2, B3, F4
Anchor tool	Issue tree (MECE)
Supporting tools (demonstrated only)	Hypothesis-led tree; The sealed working answer; The Analyst's Ledger
Theory link	None today.
Gate	No gate today.

Learning objectives

1. Apply the two MECE tests to a set of branches.
2. Build an issue tree to three levels that ends in answerable questions.
3. Distinguish a question-led tree from a hypothesis-led tree.
4. Write a sealed working answer of no more than 100 words.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 30	Concept: MECE and trees	20 minutes
30 to 50	Demonstration: Madhyapur tree	20 minutes
50 to 85	Activity: issue tree challenge	35 minutes
85 to 110	Own case: tree and sealed answer	25 minutes
110 to 120	Debrief and exit task	10 minutes

Preparation and materials

- The Day 3 slide deck (03 Slides)
- Worksheet 2.1 in Workbook 2
- Sticky notes or a shared whiteboard
- MECE test card

Indian policy context

A state scheme that gives bicycles to girls entering secondary school can be split by supply and demand: was the bicycle offered and delivered, and was it taken up? Each branch leads to different evidence and different decision-makers.

Teaching narrative, slide by slide

Slide 1. Title slide. Return provisional case approvals before you begin. Anyone whose case needs changes gets a two-line note now and a short conversation during the workbook block.

Slide 2. Today. Session outcome: by the end of the session each participant has built a three-level MECE issue tree for your case, written and sealed a working answer, and started your Analyst's Ledger.

Slide 3. Opening problem. Collect reasons on the board in the order they are called out. The list will be unordered, with overlaps (awareness and attitude) and gaps (nobody mentions processing capacity). That untidy list is what an issue tree fixes.

Slide 4. MECE: two tests for every split. Apply both tests to the opening list on the board. Show one overlap and one gap. Make the point in the note explicitly: social causes interact, so MECE applies to how we organise questions, not to how the world works.

Slide 5. A question-led issue tree for Madhyapur. Build this live if you can, branch by branch. Ask the room to test level one for overlap and gap. Someone usually asks whether enforcement belongs inside households; it does not, because it is a separate actor and lever. Point out that the leaves are questions, not solutions.

Slide 6. Four ways to split a problem. Ask which split would suit a Commissioner deciding what to fund. Stage splits are usually best for delivery problems because each stage has an owner. The driver tree is powerful but needs numbers; it will matter again on Day 12.

Slide 7. Question-led and hypothesis-led trees. Consulting practice favours hypothesis-led work because it is fast. The cost is confirmation bias. The Lab uses both: the question-led tree protects coverage, the sealed answer forces a commitment that you will test from Day 9 onwards.

Slide 8. The sealed working answer. Collect sealed answers separately, by email or in an envelope. They are not marked and not read until Day 20. Tell participants that changing their mind in response to evidence will be credited in the reflection, so they have no reason to write a cautious answer.

Slide 9. The Analyst's Ledger starts today. The Ledger is a habit, not a form. It makes analytical humility visible. At each gate the instructor samples ten claims; the standard is nine correctly tagged. Practise on two claims from the opening list.

Slide 10. Activity: The issue tree challenge. Expect first-level splits that mix logics, such as awareness, distance, family attitudes and scheme design. Ask the pod to name its logic; a mixed list cannot pass the MECE test. A supply and demand split works well here: was the bicycle offered and delivered, and was it taken up?

Slide 11. Your case: Worksheet 2.1. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Ask each participant to name their splitting logic aloud before they build level two.

Slide 12. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Name your level-one splitting logic in five words.

Slide 13. Before the next session. Show your tree to someone outside the course and ask them to find a gap. Revise it. Add one more answerable question to your weakest branch. About 40 minutes.

Likely misconceptions

Misconception	How to respond
An issue tree is a mind map.	A mind map collects associations; an issue tree divides a question into parts that do not overlap and together cover it.
More branches make a better tree.	More branches usually mean overlap. Three to five per level is typical.

Misconception	How to respond
Writing an answer early is biased.	Only if you protect it. The sealed answer exists to be tested and, often, overturned.

Discussion questions

Question	What strong answers include
Why is MECE hard for social problems?	Social causes interact, so clean separation is difficult. Strong answers say MECE disciplines the structure of questions and is not a claim that causes are independent.
When is a hypothesis-led tree better?	When there is enough knowledge for a credible answer and time is short. Strong answers name the risk of confirmation bias and how the hypothesis register on Day 9 guards against it.

Activity: The issue tree challenge

Element	Detail
Purpose	Build a MECE tree under time pressure, then defend its logic against another pod.
Scenario	Suppose a state scheme that gives bicycles to girls entering secondary school is reported to have low uptake in some districts. The state department asks why. Build the tree before you know anything else.
Group size	Pods of four
Time	35 minutes
Materials	Sticky notes or a shared whiteboard; MECE test card
Expected output	A three-level tree with its splitting logic stated

Steps.

1. Agree the question at the root in five minutes.
2. Choose a splitting logic and build level one. Test it for MECE.
3. Build two more levels until every leaf is an answerable question.
4. Swap trees with another pod. Find one overlap and one gap.
5. Repair your tree and note what changed.

Facilitation. Expect first-level splits that mix logics, such as awareness, distance, family attitudes and scheme design. Ask the pod to name its logic; a mixed list cannot pass the MECE test. A supply and demand split works well here: was the bicycle offered and delivered, and was it taken up?

Debrief questions.

- Which first level passed the MECE test without repair, and what logic did it use?
- Where did the swap find a gap?
- Which leaf would you investigate first, and why?

Evaluation criteria	Common errors
One splitting logic per level	Mixing splitting logics at one level
No overlaps or gaps at level one	Leaves that are solutions rather than questions
Leaves are answerable questions	Stopping at two levels

Extension for strong groups. Build a driver tree: uptake equals eligible girls times the share who know, times the share who apply, times the share who receive.

Workbook application: Worksheet 2.1, Issue tree and sealed working answer

Turn your problem into a structured set of answerable questions, and commit to a first answer you will test. Ask each participant to name their splitting logic aloud before they build level two.

Hand in. Worksheet 2.1 by the start of Day 4. Send the sealed answer to the instructor separately; it is not marked and not opened until Day 20.

Independent work

Show your tree to someone outside the course and ask them to find a gap. Revise it. Add one more answerable question to your weakest branch. About 40 minutes.

Assessment

Part of Gate 1 (Day 5). Feedback uses competencies B1 and B2 and tracker codes ST1 and ST2.

Readings

- Essential: Conn and McLean, Bulletproof Problem Solving (2018), the chapter on disaggregating problems with logic trees.
- Optional: Minto, The Pyramid Principle, the chapter on grouping ideas (edition to confirm).

Close and transition

Exit task: Name your level-one splitting logic in five words. Transition: Tomorrow we move from questions to causes, using problem trees, Five Whys and the fishbone.

Day 4: Finding causes

Item	Detail
Week	1: Define and structure (Thursday)
Central question	How do we move from describing a problem to explaining why it exists?
Session outcome	Each participant has built a problem tree for your case that separates symptoms from proximate and structural causes, and tested a Five Whys chain against it.
Competencies	C1, C2, C5
Anchor tool	Problem tree
Supporting tools (demonstrated only)	Five Whys; Fishbone diagram
Theory link	None today.
Gate	No gate today.

Learning objectives

1. Separate symptoms, proximate causes and structural causes.
2. State the mechanism behind each causal link.
3. Use Five Whys and the fishbone to generate candidate causes without treating them as proof.
4. Distinguish correlation, causation and mechanism.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 30	Concept: levels of cause	20 minutes
30 to 45	Demonstration: Madhyapur problem tree	15 minutes
45 to 90	Thursday challenge: root cause investigation	45 minutes
90 to 110	Own case: problem tree	20 minutes
110 to 120	Debrief and exit task	10 minutes

Preparation and materials

- The Day 4 slide deck (03 Slides)
- Worksheet 2.2 in Workbook 2
- Eight cause cards per pod
- A large sheet or shared board

Indian policy context

Foundational learning: low reading levels in early grades are commonly linked to several candidate causes at once, including teaching at grade level rather than at the child's level, multigrade classrooms, home language and attendance. A single-cause story rarely survives the evidence.

Teaching narrative, slide by slide

Slide 1. Title slide. Today is the first Thursday challenge, so the group activity runs longer. Keep the concept block tight.

Slide 2. Today. Session outcome: by the end of the session each participant has built a problem tree for your case that separates symptoms from proximate and structural causes, and tested a Five Whys chain against it.

Slide 3. Opening problem. Good answers separate the claim into an association (absence and learning move together) and a mechanism (absent teachers mean less instruction). Ask what else could explain the association: remote schools may have both more absence and lower learning for other reasons.

Slide 4. Three levels of explanation. The distinction matters for the decision. The Commissioner can fund street cleaning (symptom), a campaign (proximate) or a contract change (structural). Each has a different cost and durability. Ask participants to name one element of each type in their own case.

Slide 5. A problem tree for Madhyapur: causes and mechanisms. Read the tree from right to left: deeper causes produce mechanisms that produce the core problem. Every mechanism is still a hypothesis. Ask which of the four branches the Commissioner controls most directly. The contract branch stands out, which is why it will reappear in the hypotheses on Day 9.

Slide 6. Five Whys, used carefully. Five Whys comes from Toyota's production system. Its strength is persistence; its weakness is that it follows one path and stops where the analyst's beliefs stop. Run a second chain from a different starting point (households lack space) to show a different structural cause.

Slide 7. A fishbone generates candidates across categories. The fishbone, from Kaoru Ishikawa's quality work, forces the analyst to look across categories, which protects against fixating on one. Its weakness is that it lists without ranking or linking. Use it to generate, then move candidates onto the problem tree.

Slide 8. Root-cause tools: where they mislead. The last row matters in public policy. Explanations that end with citizens not caring are usually incomplete: ask what makes the behaviour rational given the costs and signals people face.

Slide 9. Correlation, causation and mechanism. This distinction returns on Day 12 with data. For now the habit is linguistic: participants should write moves together, is associated with, or causes, and mean the difference.

Slide 10. Activity: Root cause investigation. The common trap is to place low parental literacy at the root, which puts responsibility outside the school. Ask what the school system controls. The teaching-at-grade-level card is a useful test: is it a cause, or a solution in disguise?

Slide 11. Your case: Worksheet 2.2. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Look for trees with a single cause; ask what else would have to be true for that cause to be the whole story.

Slide 12. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Write down the causal link in your tree that you are least sure of.

Slide 13. Before the next session. Revise your problem tree after the challenge. For your two most important causal links, write what evidence would show the link to be wrong. About 45 minutes.

Likely misconceptions

Misconception	How to respond
The root cause is the deepest one.	It is the one that, if changed, would most change the outcome and that someone can act on. The deepest cause may lie beyond any decision-maker's reach.

Misconception	How to respond
Five Whys finds the root cause.	It finds one chain. Different starting points give different chains.
Household background explains outcomes, so the system is not at fault.	Background matters, but the analysis should ask what the system controls and how it responds to background.

Discussion questions

Question	What strong answers include
When is a cause structural rather than proximate?	When it shapes many proximate causes and persists over time. Strong answers give an institutional example such as contract terms or staffing rules.
How would you know a causal link is wrong?	If the cause is present and the effect absent, or the reverse, across comparable cases. Strong answers name a comparison they could make with published data.

Activity: Root cause investigation

Element	Detail
Purpose	Separate symptoms, proximate causes and structural causes in an unfamiliar case, and defend the links.
Scenario	In a stylised district, many children in government primary schools cannot read a simple paragraph by the end of Grade 3. Each pod gets eight cards: teacher vacancies; multigrade classrooms; home language differs from the textbook language; irregular attendance at harvest time; assessment focused on finishing the syllabus; low parental literacy; frequent non-teaching duties for teachers; children taught at grade level rather than at their learning level.
Group size	Pods of four
Time	45 minutes (Thursday challenge)
Materials	Eight cause cards per pod; A large sheet or shared board
Expected output	A problem tree with mechanisms and one challenged link

Steps.

1. Sort the eight cards into symptoms, proximate causes and structural causes.
2. Build a problem tree, writing the mechanism on each arrow.
3. Run one Five Whys chain from the card you trust least.
4. Present your tree in two minutes; another pod challenges one link.
5. Mark which links you would test first, and why.

Facilitation. The common trap is to place low parental literacy at the root, which puts responsibility outside the school. Ask what the school system controls. The teaching-at-grade-level card is a useful test: is it a cause, or a solution in disguise?

Debrief questions.

- Which card was hardest to place, and why?
- Did any pod treat a solution as a cause?
- Which link would most change the recommendation if it proved false?

Evaluation criteria	Common errors
Each element placed at the right level	Single-cause trees
A mechanism on every arrow	Arrows without mechanisms
At least one challenged link answered	Treating household background as the end of the explanation

Extension for strong groups. Show how the tree would differ for a decision-maker at state level rather than district level.

Workbook application: Worksheet 2.2, Problem tree for your case

Explain why your problem exists, separating levels of cause and stating the mechanism behind each link. Look for trees with a single cause; ask what else would have to be true for that cause to be the whole story.

Hand in. Worksheet 2.2 by the start of Day 5, for inclusion in Gate 1.

Independent work

Revise your problem tree after the challenge. For your two most important causal links, write what evidence would show the link to be wrong. About 45 minutes.

Assessment

Part of Gate 1. Feedback uses competencies C1, C2 and C5 and tracker codes CA1 and CA2.

Readings

- Essential: Andrews, Pritchett and Woolcock, Building State Capability (2017), the chapter on problem deconstruction, which uses the fishbone diagram (open access).
- Optional: Pratham, Annual Status of Education Report: the summary of the most recent national report, for the common case.

Close and transition

Exit task: Write down the causal link in your tree that you are least sure of. Transition: Tomorrow we look for the feedback loops that keep your problem in place, and lock your scope at Gate 1.

Day 5: Systems and scope lock

Item	Detail
Week	1: Define and structure (Friday)
Central question	What keeps the problem in place, and is our case now ready for analysis?
Session outcome	Each participant has drawn a causal loop diagram for your case, checked the obvious fix for unintended consequences, and submitted Gate 1.
Competencies	C3, C4, B2
Anchor tool	Causal loop diagram
Supporting tools (demonstrated only)	Unintended consequences check; Gate 1 pod peer review
Theory link	Andrews, Pritchett and Woolcock: Would the obvious fix copy the form of a programme that worked elsewhere without the capability behind it?
Gate	Gate 1 is due today.

Learning objectives

1. Identify a reinforcing or balancing loop that sustains the problem.
2. Mark the polarity of each link.
3. Anticipate at least two unintended consequences of the obvious fix.
4. Lock the scope of your case and submit the problem diagnostic.

Session plan

Minutes	Block	Length
0 to 10	Opening problem	10 minutes
10 to 25	Concept: loops and polarity	15 minutes
25 to 40	Demonstration: Madhyapur loop	15 minutes
40 to 65	Activity: unintended consequences	25 minutes
65 to 105	Gate 1: pod peer review	40 minutes
105 to 120	Debrief and exit task	15 minutes

Preparation and materials

- The Day 5 slide deck (03 Slides)
- Worksheet 2.3 in Workbook 2
- Blank loop template
- Actor cards: vendors, customers, police, shopkeepers, commuters

Indian policy context

Banning street vending on main roads to ease congestion can push vendors into side streets, raise informal payments and move the congestion. The Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, 2014 makes the design of such measures a legal as well as a practical question.

Teaching narrative, slide by slide

Slide 1. Title slide. Gate day. The concept block is shorter to make room for 40 minutes of pod peer review. Remind participants that the Gate 1 submission is due at the end of the day.

Slide 2. Today. Session outcome: by the end of the session each participant has drawn a causal loop diagram for your case, checked the obvious fix for unintended consequences, and submitted Gate 1.

Slide 3. Opening problem. Expect: households avoid the fine by dumping at night, inspectors cannot attribute roadside waste, so fines are rarely issued and lose credibility. This is a balancing response that defeats the policy. Keep the sketch on the board for the next slides.

Slide 4. Linear and systems explanations. Linear thinking is not wrong; it is incomplete. Most public problems that persist despite repeated interventions are held in place by a feedback loop. Finding it often explains why previous fixes failed.

Slide 5. A reinforcing loop in Madhyapur. Trace the loop aloud twice, once as a vicious circle and once as a virtuous one: if crews keep streams apart, households see it, separate more, and separated collection becomes routine. Ask which link the Commissioner can break most cheaply. This is why the contract appears in every week of the spine case.

Slide 6. Two kinds of loop, and how to mark them. Polarity is where reasoning errors hide. Work one link with the room: more fines lead to more roadside dumping (plus), more dumping leads to fewer enforceable fines (minus). A loop with an odd number of minus links is balancing.

Slide 7. Checking the obvious fix for unintended consequences. None of these effects means the fix is wrong. They mean the design must anticipate responses. The fourth row is important for the spine case, because the contract change is the likely recommendation; its transition risk will return on Day 18.

Slide 8. A thinker for your case. Andrews, Pritchett and Woolcock, Building State Capability, 2017. They describe isomorphic mimicry: governments adopting the outward form of successful reforms without the capability to make them work, which can trap a system in failure. Put the question to the room: Would the obvious fix copy the form of a programme that worked elsewhere without the capability behind it? Then ask each participant to answer it for their own case in one sentence.

Slide 9. Activity: The unintended consequences challenge. Expect vendors to move to side streets, informal payments to rise and customers to follow the vendors. Steer pods to identify who bears the cost. If participants raise the Street Vendors Act, 2014, note that it requires town vending committees and vending zones, which changes the design space.

Slide 10. Gate 1 pod peer review: 40 minutes. Circulate and listen for generic feedback such as good structure. Push reviewers to name a specific branch, link or word. The revision made in response is credited later through the revision note (competency J4).

Slide 11. Your case: Worksheet 2.3. Individual application, about 20 minutes. Circulate and check that every participant is working on their own case. Most participants will draw a chain, not a loop. Ask: what does the last box do to the first?

Slide 12. What to take away. Ask two participants to state the takeaway they found least obvious, then close with the exit task: Write the one link in your loop that your decision-maker could actually break.

Slide 13. Before the next session. Catch up and rest. If you have time, identify the institutions responsible for your problem: which ministry, state department and local body are involved? Bring notes to Day 6. About 30 minutes.

Likely misconceptions

Misconception	How to respond
Systems thinking means everything is connected, so nothing can be prioritised.	The point is to find the few loops that matter most, then act on them.
A loop diagram needs dozens of variables.	One clear loop with three to five links is more useful than a map nobody can read.
Unintended consequences mean the policy is bad.	They mean the design must anticipate responses. Many good policies have side effects that can be managed.

Discussion questions

Question	What strong answers include
Why are reinforcing loops dangerous for reformers?	They lock problems in and can turn small changes into large ones, in either direction. Strong answers find the loop in their own case and say where a small change could reverse it.
What would isomorphic mimicry look like in your case?	Adopting the form of a successful reform, such as a dashboard or a new unit, without the capability to make it work. Strong answers explain how they would tell form from function.

Activity: The unintended consequences challenge

Element	Detail
Purpose	Find how a sensible-looking intervention could make things worse, and why.
Scenario	A city government proposes to ban street vending on main roads to reduce congestion. Play out what might happen over twelve months for vendors, customers, traffic police, nearby shops and the roads themselves.
Group size	Pods of four
Time	25 minutes
Materials	Blank loop template; Actor cards: vendors, customers, police, shopkeepers, commuters
Expected output	A loop showing one unintended consequence, and a design fix

Steps.

1. Draw the intended chain: ban, fewer vendors, less congestion.
2. Add at least three actors' responses the planners did not intend.
3. Find one loop: a response that feeds back on the original problem.
4. Mark each link plus or minus.
5. Propose one design change that weakens the harmful loop.

Facilitation. Expect vendors to move to side streets, informal payments to rise and customers to follow the vendors. Steer pods to identify who bears the cost. If participants raise the Street Vendors Act, 2014, note that it requires town vending committees and vending zones, which changes the design space.

Debrief questions.

- Which unintended effect was most likely, and which most harmful?
- Whose behaviour did the planners not model?
- Does your design fix create a loop of its own?

Evaluation criteria	Common errors
At least three actor responses	Listing effects without mechanisms
One loop with polarities	Ignoring who bears the cost
A fix that addresses the mechanism	Fixes that restate the original policy

Extension for strong groups. Apply the same check to the obvious fix in your own case.

Workbook application: Worksheet 2.3, Causal loop and the Gate 1 diagnostic

Show what keeps your problem in place, anticipate the side effects of the obvious fix, and submit your revised problem diagnostic. Most participants will draw a chain, not a loop. Ask: what does the last box do to the first?

Hand in. Gate 1 diagnostic (Worksheets 1.2, 2.1, 2.2 and 2.3, revised) by the end of Day 5. Feedback returned by Day 7.

Independent work

Catch up and rest. If you have time, identify the institutions responsible for your problem: which ministry, state department and local body are involved? Bring notes to Day 6. About 30 minutes.

Assessment

Gate 1: detailed feedback on domains A, B and C under the five feedback headings. Formative; patterns recorded in the Analyst Development Tracker.

Readings

- Essential: Andrews, Pritchett and Woolcock, Building State Capability (2017), the chapter on capability traps and isomorphic mimicry (open access).
- Optional: Meadows, Donella H. Thinking in Systems: A Primer (2008), chapter 1.

Close and transition

Exit task: Write the one link in your loop that your decision-maker could actually break. Transition: Week 2 begins with jurisdiction, money and the delivery chain: who can act on your problem, and with what?